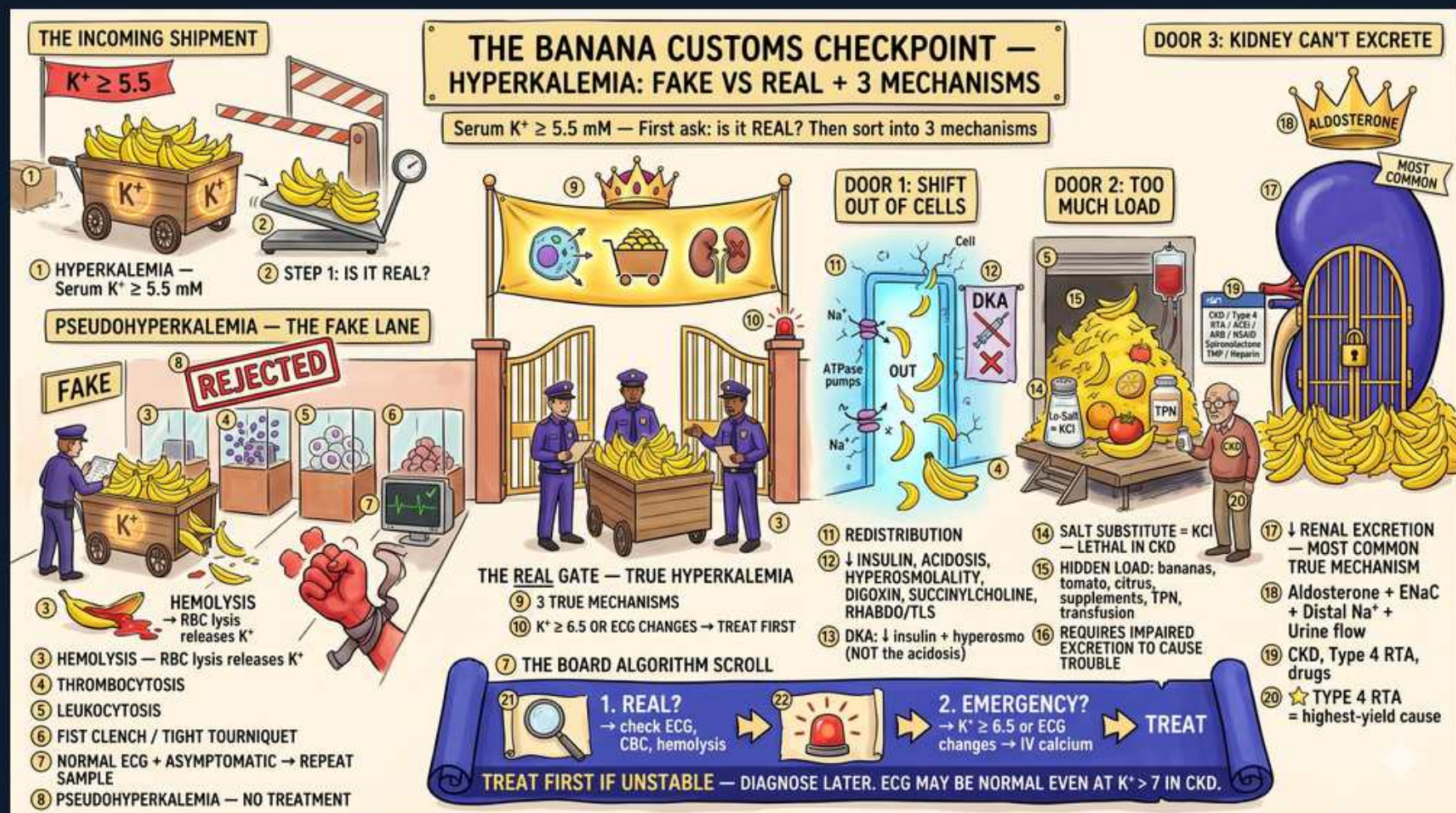


# Hyperkalemia — mechanisms (real vs fake + 3 causes)





## 1. K+ >= 5.5

### WHAT YOU SEE

Incoming banana shipment, K+ >= 5.5 sign

### WHAT IT ENCODES

Hyperkalemia is defined as serum K+ >= 5.5 mM (normal 3.5-5.0). Severity tiers: mild 5.5-6.0, moderate 6.1-6.9, severe >= 7.0 or any ECG change. The cell stores 98% of body potassium, so serum K+ is a small, tightly regulated window onto a large intracellular pool.

### BOARD TRAP

WRONG to equate a given serum K+ with total-body potassium status. Discriminator: serum K+ can be high while total body K+ is depleted (DKA) or normal (pseudohyperkalemia) — interpret the number alongside acid-base status, glucose, and how the sample was drawn.

## 2. Is it REAL?

### WHAT YOU SEE

Step 1 sign at the checkpoint

### WHAT IT ENCODES

The first question in any hyperkalemia case is whether the value is real or pseudohyperkalemia (spurious in-vitro K+ release). Reflexively confirm with a properly drawn repeat and review the sample (hemolysis note, platelet/WBC count, draw technique) if the patient is hemodynamically stable. This step prevents unnecessary, potentially harmful treatment.

### BOARD TRAP

WRONG ANSWER: rushing to insulin/calcium/kayexalate for a high K+ in a stable, asymptomatic patient with a normal ECG and a hemolyzed sample. Discriminator: if stable with a normal ECG and a suspicious draw, repeat first; if unstable or ECG changes are present, treat empirically and do not wait.

## 3. Pseudo — fake lane

### WHAT YOU SEE

FAKE / REJECTED lane

### WHAT IT ENCODES

Pseudohyperkalemia is a spurious elevation from K+ leaking out of cells after the blood is drawn; total body and in-vivo K+ are normal and no treatment is needed. Clues: a hemolysis flag, marked thrombocytosis/leukocytosis, difficult/traumatic draw, or a normal ECG that doesn't fit the number. Confirm with a fresh, atraumatic plasma sample.

### BOARD TRAP

WRONG to treat pseudohyperkalemia as a true emergency. Discriminator: pseudohyperkalemia has NO ECG changes and a benign clinical picture — recognizing it avoids iatrogenic hypokalemia; treating it can drive the real K+ dangerously low.

## 4. Hemolysis

### WHAT YOU SEE

Burst red blood cells releasing K+

### WHAT IT ENCODES

In-vitro hemolysis — from a small-gauge needle, excessive aspiration, or delayed/agitated processing — lyses RBCs and leaks K+ into the serum, falsely raising the value. The lab often flags the specimen as hemolyzed. The fix is a clean, atraumatic redraw, not treatment.

### BOARD TRAP

WRONG to act on a hemolyzed sample's K+. Discriminator: a hemolysis flag plus a normal ECG points to a spurious result — redraw with good technique; do not confuse in-vitro hemolysis with in-vivo hemolysis (which is true hyperkalemia with high LDH, low haptoglobin, elevated indirect bilirubin).

## 5. Thrombocytosis

### WHAT YOU SEE

Clumped platelets

### WHAT IT ENCODES

Platelets and leukocytes release potassium as blood clots in the tube, so marked thrombocytosis or leukocytosis falsely elevates SERUM K+ while PLASMA K+ stays normal. This is reverse pseudohyperkalemia's classic counterpart. Drawing a heparinized plasma (unclotted) sample reveals the true value.

### BOARD TRAP

WRONG to diagnose true hyperkalemia from a serum sample in a patient with extreme thrombocytosis/leukocytosis (e.g., myeloproliferative disease). Discriminator: serum (clotted) K+ is high but plasma (unclotted) K+ is normal — order a plasma potassium to unmask the artifact.

## 6. Fist/tourniquet

### WHAT YOU SEE

Clenched fist / tight tourniquet

### WHAT IT ENCODES

Repeated fist clenching or a prolonged/tight tourniquet causes local muscle K+ release and forearm ischemia, spuriously raising the drawn K+. Proper technique (release tourniquet, avoid pumping the fist) prevents it. This is a pure preanalytic artifact.

### BOARD TRAP

WRONG to treat a mildly elevated K+ from a tourniquet-and-fist-pump draw in a well patient. Discriminator: technique-related pseudohyperkalemia resolves on a properly drawn repeat — fix the draw, not the patient.

## 7. Door 1: shift

### WHAT YOU SEE

Door 1 — K+ shifting OUT of a cell

### WHAT IT ENCODES

Transcellular shift = potassium moves OUT of cells into the extracellular space without a change in total body K+. Drivers: acidosis (especially mineral acids), insulin deficiency, hyperosmolality, beta-blockade, digoxin (Na/K-ATPase inhibition), succinylcholine, and cell lysis. Therapy that pushes K+ back IN (insulin/glucose, beta-2 agonists) exploits the reverse of these mechanisms.

### BOARD TRAP

WRONG to assume a shift-driven hyperkalemia reflects potassium excess that must be removed. Discriminator: in pure shift, total body K+ is normal (or low, as in DKA) — treat the underlying driver and shift K+ back into cells; aggressive removal can cause rebound hypokalemia once the shift reverses.



## 8. DKA shift

### WHAT YOU SEE

Banana with DKA label leaving cell, ATPase pump

### WHAT IT ENCODES

In DKA/HHS, insulin deficiency plus hyperosmolality (and solvent drag) drives K<sup>+</sup> out of cells, so serum K<sup>+</sup> may be high even though total body K<sup>+</sup> is markedly DEPLETED from osmotic diuresis. As insulin and fluids are given, K<sup>+</sup> shifts back into cells and serum K<sup>+</sup> falls fast. Note: DKA's anion-gap (ketoacid) acidosis itself shifts K<sup>+</sup> less than mineral-acid acidosis does — the dominant drivers are insulin deficiency and hyperosmolality.

### BOARD TRAP

WRONG ANSWER: withholding potassium in DKA because the initial serum K<sup>+</sup> looks normal or high. Discriminator: total body K<sup>+</sup> is depleted — once serum K<sup>+</sup> falls below ~5.2-5.3 mM, add K<sup>+</sup> to fluids, and HOLD insulin if K<sup>+</sup> < 3.3 mM until repleted to avoid lethal hypokalemia.

## 9. Door 2: load

### WHAT YOU SEE

Door 2 — too much load, pile of bananas

### WHAT IT ENCODES

Excess potassium load (exogenous: diet, salt substitutes, supplements, KCl, transfusion; endogenous: TLS, rhabdomyolysis, hemolysis) causes hyperkalemia chiefly when excretion is impaired. Healthy kidneys clear large loads, so load alone is rarely sufficient. Endogenous lysis loads are especially dangerous because they are large and ongoing.

### BOARD TRAP

WRONG to blame dietary/load causes when GFR and aldosterone are normal. Discriminator: with intact kidneys, load alone rarely sustains hyperkalemia — search for a concurrent excretion defect (CKD, hypoaldosteronism, drugs) that converts the load into a problem.

## 10. Salt sub = KCl

### WHAT YOU SEE

Salt substitute container

### WHAT IT ENCODES

Salt substitutes and 'lite'/low-sodium salts are potassium chloride — a hidden, frequently underreported load. In CKD or on RAAS blockade they readily precipitate severe hyperkalemia. Always ask specifically about salt substitutes, supplements, herbals, and recent transfusions.

### BOARD TRAP

WRONG to miss the salt substitute because the patient denies taking 'potassium.' Discriminator: patients view salt substitutes as food, not medicine — the classic deadly setup is elderly + DM + CKD on an ACEi who switched to 'lite' salt.

## 11. Door 3: kidney

### WHAT YOU SEE

Door 3 — kidney can't excrete, aldosterone crown

### WHAT IT ENCODES

Decreased renal excretion is the MOST COMMON cause of true, sustained hyperkalemia: reduced GFR (CKD/AKI), low aldosterone effect (hypoaldosteronism, Type 4 RTA, Addison's), and drugs (ACEi/ARB, MRAs, ENaC blockers, NSAIDs, trimethoprim, heparin, calcineurin inhibitors). Effective secretion needs aldosterone, ENaC, distal Na<sup>+</sup> delivery, and urine flow — failure of any limits excretion. This is the highest-yield bucket for chronic/recurrent hyperkalemia.

### BOARD TRAP

WRONG to anchor on shift or load and overlook an excretion defect in chronic/recurrent hyperkalemia. Discriminator: persistent hyperkalemia almost always involves impaired renal excretion — assess GFR, urine output, aldosterone status, and a medication list before settling on a shift/load cause.

## 12. Type 4 RTA

### WHAT YOU SEE

Type 4 RTA highlighted as highest-yield

### WHAT IT ENCODES

Type 4 RTA = hyporeninemic hypoaldosteronism, the highest-yield decreased-excretion cause, classically in diabetic nephropathy. It produces a NORMAL-anion-gap (hyperchloremic) metabolic acidosis WITH hyperkalemia and an inappropriately acidic-to-mildly-impaired urine. Treat by removing offending drugs, low-K<sup>+</sup> diet, loop/thiazide diuretics, and fludrocortisone in selected patients.

### BOARD TRAP

WRONG to confuse Type 4 with Type 1 (distal) RTA. Discriminator: Type 4 RTA is the ONLY RTA with HYPERkalemia (Types 1 and 2 cause HYPOkalemia) — high K<sup>+</sup> plus NAGMA in a diabetic points straight to hyporeninemic hypoaldosteronism.

## 13. Emergency? ECG

### WHAT YOU SEE

Board algorithm scroll: REAL? -> EMERGENCY?

### WHAT IT ENCODES

After confirming the value is real, assess for emergency: K<sup>+</sup> >= 6.5 mM OR any ECG change triggers immediate IV calcium for membrane stabilization, then shift (insulin/glucose, albuterol) and removal (dialysis, loop diuretics, GI binders). The ECG, not the number alone, governs urgency. Get a STAT ECG and monitor early.

### BOARD TRAP

WRONG to rely on the absolute K<sup>+</sup> value and be reassured by a normal ECG in CKD/ESRD. Discriminator: the ECG can stay NORMAL even at K<sup>+</sup> > 7 in chronically adapted CKD/ESRD patients — treat based on the trend, rate of rise, and clinical context, not solely the tracing.

## 14. Treat if unstable

### WHAT YOU SEE

TREAT banner at bottom

### WHAT IT ENCODES

If the patient is unstable or shows ECG changes, treat first and diagnose later — stabilize the membrane with IV calcium, shift K<sup>+</sup> intracellularly, and remove it, before completing the full etiologic workup. Mnemonic 'C BIG K Drop': Calcium, Bicarb (if acidotic), Insulin/glucose, Glucose, Kayexalate/binders, Dialysis/Diuretics. Definitive cause-finding follows stabilization.

### BOARD TRAP

WRONG to delay treatment to obtain aldosterone levels, TTKG, or other workup in an unstable hyperkalemic patient. Discriminator: hemodynamic instability or ECG changes mandate empiric treatment now — the diagnostic workup is completed after the emergency is controlled, never before.